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Nonclinical Considerations for Support of Pregnancy and Lactation in Clinical Trials

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ABSTRACT

Background: Pregnant and breastfeeding individuals remain underrepresented in clinical drug development, leaving health care providers and patients with limited data to inform treatment decisions. Current practice relies heavily on nonclinical developmental and reproductive toxicity (DART) studies in animals, while clinical trials typically exclude pregnant and breastfeeding populations or require highly effective contraception for women of childbearing potential. Consequently, medications may be used during pregnancy or lactation despite limited supporting evidence, which can lead to treatment interruption, suboptimal dosing, and increased maternal or fetal risk. Emerging regulatory guidance, including draft ICH Guideline on Inclusion of Pregnant and Breastfeeding Individuals in Clinical Trials (E21), highlights the ethical and clinical need to generate data directly in these populations.

Methods: This paper evaluates the current regulatory framework for nonclinical studies described in ICH S5(R3) and M3(R2) and examines how these studies can support the enrollment of pregnant and breastfeeding individuals in clinical trials. It proposes pragmatic adaptations to align nonclinical testing with the anticipated clinical context of use and considers the role of new approach methodologies (NAMs) and alternative study designs.

Results: Potential DART strategies are described that could enable the inclusion of pregnant individuals in clinical trials based on factors such as dosing duration, route of administration, and study type (e.g., pharmacokinetic studies). The paper also considers the use of NAMs, including physiologically based pharmacokinetic modeling and in vitro assays, as well as alternative study designs

Abbreviations: DART, developmental and reproductive toxicity; EFD, embryo-fetal development; ePPND, enhanced pre- and postnatal development; FcRn, neonatal Fc receptor; FDA, Food and Drug Administration; FEED, fertility and early embryonic development; GD, gestational day; GW, gestational week; ICH, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use; mAbs, monoclonal antibodies; NCE, new chemical entity; NHP, nonhuman primate; PBPK, physiologically based pharmacokinetics; PK, pharmacokinetics; popPK, population pharmacokinetics; PPND, pre- and postnatal development; WOCBP, women of child-bearing potential; WoE, weight of evidence.

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such as preliminary pre- and postnatal development or extended embryo–fetal development studies. Approaches to inform studies in breastfeeding populations, including milk–plasma ratio modeling and translation from nonclinical data, are also discussed.

Conclusions: These approaches support a shift from routine exclusion toward proactive and ethical inclusion of pregnant and breastfeeding individuals in clinical research, helping address longstanding evidence gaps and improving therapeutic decision-making.

1 | Introduction

There is a growing recognition of the need for clinical data to inform the use of medicinal products in pregnant and breastfeeding individuals (in the context of this paper, and in alignment with draft ICH E21, “individuals” refers to people who have the potential to become pregnant) (Harrilal Khan et al. 2025; National Academies of Sciences 2024; PRGLAC 2018). Without data, health care professionals and patients struggle to assess risks and benefits, complicating appropriate care. For example, in the United States, as many as 70% of women took at least one prescription medication during pregnancy (Mitchell et al. 2011). Lack of medicinal product information on efficacy and safety during pregnancy may lead to avoidance of treatment, under-treatment, or discontinuation upon pregnancy detection even if the medication is beneficial and potentially safe during pregnancy. As a result, risks of morbidity and mortality from acute or chronic illnesses may be increased during pregnancy and breastfeeding. In 2024, the National Academy of Sciences, Engineering and Medicine authored a report that proposed nine recommendations to the US Congress that would promote research in pregnant and lactating women to reduce this data gap (National Academies of Sciences 2024). One of these recommendations included revision of the FDA guidelines to mandate initiation of trials in pregnant and lactating women no later than the end of Phase 3, i.e., before product registration, for any drugs in development for the general population.

For the majority of medicinal products, the information in the drug product labeling related to the risks during pregnancy and breastfeeding is obtained from nonclinical developmental and reproductive toxicity (DART) studies in animals, with limited human data, if any, at the time of market authorization. As a result, the prescriber must translate nonclinical safety information into a relevant human risk to determine the overall benefit vs. risk to a pregnant or breastfeeding patient when making prescribing decisions. However, if pregnant and breastfeeding individuals were to be included in preregistrational trials, with appropriate informed consent and proper safeguards, drug developers could generate data on pharmacokinetics (PK), efficacy and safety that could inform on the appropriate dose and the safety profile in pregnant and breastfeeding individuals, enabling an improved assessment of benefit and risk. Having this information on a product label would help to provide more relevant information to the health care provider, which would ultimately benefit a pregnant or breastfeeding patient. To this end, the ethically responsible inclusion of pregnant individuals in clinical trials has been proposed with the assumption that it is “generally safer for pregnant individuals to use drugs within a trial under controlled conditions with proper follow-up than to take drugs that are prescribed” without having been studied in this population (Baylis 2010).

Therefore, drug development plans for pharmaceuticals that may benefit pregnant or lactating individuals need to incorporate

strategies that enable the enrollment of these individuals into clinical trials prior to registration of those pharmaceuticals. To achieve this goal, early clinical trials (e.g., Phase 1) need to include women of childbearing potential (WOCBP) as early in development as possible. This will enable subsequent clinical trials to obtain clinical efficacy and safety data in WOCBP to support a favorable prospect of benefit over risk. For drug development teams, this would mean the conduct of nonclinical DART studies as soon as possible to support their enrollment. Once drug development teams have sufficient data on clinical safety in WOCBP, a good understanding of dose/exposure vs. efficacy relationship (i.e., a prospect of benefit) along with the DART data that demonstrate low or negligible risk during pregnancy, such a comprehensive data package could mark an important inflection point that demonstrates an overall favorable prospect of benefit over potential risk to pregnant participants. Once this inflection point is reached, the drug development team could move ahead with removal of mandatory contraception requirements, permit the retention of individuals who become pregnant in clinical trials, and enroll pregnant women into clinical trials. The types of clinical trials where this strategy can be employed depend on the target indication, and could include both pregnant PK trials, proof of concept trials, or registration trials. In addition, anticipating when the inflection point could be reached in development and having evidence-based flexible clinical protocol wording with adaptive features could reduce complexity of clinical trial management. This type of approach is recommended and has been followed by drug development teams developing therapeutics for HIV (Bekker et al. 2025) and malaria (El Gaaloul et al. 2022). More specifically, the WHO and International Maternal Pediatric Adolescent AIDS Clinical Trials (IMPAACT) Network recently proposed that drug developers conduct necessary nonclinical studies as soon as possible to enable the enrollment of WOCBP into clinical trials and to eventually progress into conducting clinical trials in women without requirements for contraception and retaining individuals who become pregnant in these trials with ongoing treatment and pregnancy follow-up, including conducting trials in pregnant individuals (Penazzato et al. 2022). For drugs to treat pregnancy-related conditions (e.g., preterm labor, gestational diabetes, preeclampsia), data collected during pregnancy needs to be obtained as a condition of approval in order to demonstrate efficacy and document safety/tolerability profile and, therefore, these drugs would also require a specific strategy for how to initiate treatment in pregnant individuals as soon and as safely as possible.

In order to implement the clinical strategy outlined above, a robust nonclinical risk assessment strategy needs to be developed that considers the timing of nonclinical studies in relation to the enrollment of specific types of study participants (e.g., WOCBP, pregnant and/or lactating individuals). The scope of this paper is to explore the current regulatory landscape and discuss the types of DART studies that are currently expected to support enrollment of pregnant and lactating individuals in clinical trials, as this is a patient population that is greatly understudied and

underserved with the currently available medications. It will also explore the types of DART studies that would be considered pragmatic based on duration of intended clinical exposure, intended trimester of exposure (e.g., 1st vs. 3rd trimester), and route of exposure to support clinical trials in pregnant and lactating individuals. Finally, this paper will explore whether the DART study designs and their timing within the framework of ICH S5(R3) and in M3(R2) could be adapted to obtain early data to accelerate the clinical strategy for certain indications to optimize development efficiency and relevance. The focus of this paper is on nonpregnancy specific indications, but principles within the paper could also be extrapolated to drugs intended for pregnancy-specific conditions as well.

2 | Current Regulatory Landscape

The designs of the nonclinical DART studies are described in the ICH S5(R3) guideline and the timing for these studies in relation to enrollment of specific clinical trial populations during clinical development is described in ICH M3(R2) (ICH 2009, 2020a). In general, for new chemical entities (NCEs), three separate study designs are conducted to assess DART: the fertility and early embryonic development (FEED) study, embryo-fetal development (EFD) study in two species (rodent and nonrodent), and pre- and postnatal development (PPND) study.

Based on ICH M3, in certain circumstances WOCBP can be enrolled without completed EFD studies; for example, in well-controlled short duration trials of up to 2 weeks with appropriate precautions to prevent pregnancy exposures (highly effective contraception and pregnancy tests) and specific Informed Consent Documents highlighting the absence of EFD data. This approach can be for example implemented in a very First-In-Human study where test drugs are administered in a cautious single-dose escalation scheme followed by a short multiple dose administration. Another circumstance could be where there is a predominance of the disease in women, and the objectives of the clinical trial cannot be effectively met without inclusion of WOCBP and there are sufficient precautions to prevent pregnancy. Additional considerations for the conduct of studies in WOCBP without the nonclinical developmental toxicity studies include the conduct of a weight of evidence (WoE) evaluation that includes the knowledge of the mechanism of action of the agent, the type of pharmaceutical agent, the extent of maternal/fetal exposure or the difficulty of conducting developmental toxicity studies in an appropriate animal model. Except for the circumstances outlined above, some form of EFD studies (preliminary or full-scale) is expected to be completed prior to the inclusion of WOCBP in clinical trials in non-US regions. In the US, WOCBP can be included in Phase 1 and 2 studies without EFD studies, provided there are steps taken for prevention of pregnancy (i.e., frequent pregnancy testing and highly effective contraception). EFD studies and the standard battery of genotoxicity tests are needed prior to any clinical trials that recruit WOCBP not using highly effective contraception or whose pregnancy status is unknown (HMA 2024; ICH 2009, 2009b). Both EFD and FEED studies are generally needed for initiation of Phase 3 or large-scale clinical trials in the target patient population (ICH M3(R2)). For highly specific biopharmaceuticals such as monoclonal antibodies (mAbs) that can only be tested in a nonhuman

primate (NHP), an enhanced pre- and postnatal (ePPND) study (ICH S6, 2011) can be used to support the inclusion of WOCBP. As these studies have long durations in line with mAbs pharmacokinetics, the current guidelines allow these studies to be conducted to support registration as long as pregnancy exposure precautions are in place during the conduct of the clinical trials. However, when preventing pregnancy exposure is not possible, the use of interim reports that describe the pregnancy outcomes (including assessment of infant viability and external morphological assessment) to a biopharmaceutical can support inclusion of WOCBP in a Phase 3 clinical trial (ICH 2011; Pentsuk and van der Laan 2009). For drugs indicated for advanced cancer developed under the ICH S9 guideline (ICH 2009a, 2009c) DART studies are typically limited to EFD(s) that are completed to support registration or the assessment of embryo-fetal development risk could be conducted via a WoE when adverse pregnancy outcomes are expected. Further, use of these drugs in patients, regardless of pregnancy status, is primarily driven by a risk/benefit analysis given the severity of disease. Expected risk of adverse pregnancy outcomes is mitigated by pregnancy prevention recommendations or by administering drugs to pregnant subjects outside the window of greatest susceptibility for fetal malformations (i.e., 2nd or 3rd trimester).

In female nonclinical species, the FEED study assesses the effect of a drug on reproductive cycles, ovulation, implantation, and early embryonic development. From the perspective of clinical trials that do not require mandatory contraception, adverse impact to these biological processes would generally manifest as altered menstrual cycles, infertility, or miscarriage in study participants. Therefore, the FEED studies have the ability to assess the risk of miscarriage during early pregnancy for clinical trials that do not require mandatory contraception; however, these studies are less relevant for drugs intended for use at a later pregnancy stage. The EFD study evaluates the effect of a drug from implantation to the end of organogenesis, and any adverse impact on these processes could lead to pregnancy loss or birth defects in humans. The PPND study, when conducted in rodents and rabbits, evaluates the effect of a drug from implantation through birth to sexual maturity, whereas ePPND studies in NHPs are limited in their scope of assessments and evaluate pregnancy outcomes and postnatal growth and development in a single study. What differentiates a pre-postnatal development study (rodent/rabbit or NHP) from the EFD study is that it provides information on the impact of a drug on fetal growth, parturition, lactation, and the impact on postnatal growth and development of offspring that has been exposed during pregnancy and lactation. From a human risk assessment perspective, both the PPND and the ePPND studies provide critical information on developmental outcomes of prenatal and lactational exposure to a potential therapeutic.

In a typical drug development program, almost all clinical trials implement mandatory contraception requirements and exclude pregnant individuals, even when information from nonclinical DART studies (primarily EFD studies) is available. The nonclinical DART studies help to inform the investigators and trial subjects of the potential risks of an inadvertent pregnancy exposure and the level of rigor necessary to prevent pregnancy exposure (e.g., frequency of pregnancy testing, types of contraception, etc.). However, at present, if

a pregnancy does occur during a clinical trial, dosing is generally halted, the subject is removed from the trial to prevent further exposure, and the outcome of pregnancy is monitored. As such, most inadvertent pregnancy exposures are limited to the preimplantation to late 1st to early 2nd trimester period, except for long-acting oral or injectable therapeutics. Under this current scenario, the EFD and FEED studies are generally sufficient for clinical trial management, while PPND/ePPND studies are not needed until registration, because pregnant women generally do not continue to be dosed once the pregnancy is identified. The totality of DART study data is, therefore, not available until the time of registration and it is only at this point that there is sufficient nonclinical and clinical safety and efficacy data to provide the complete benefit-risk assessment on the use of the drug in WOCBP and what precautions, if any, are needed to protect pregnant individuals. It should be noted that postauthorization a drug could be used in pregnant individuals even if it has not been studied, as long as it is not contraindicated for use during pregnancy.

Because of the current paradigm, where the totality of DART data is not required until the time of registration, a complete picture of the safety of the drug during pregnancy is typically not available when clinical trials are initiated in a drug development program, and that may be an impediment to retaining participants in clinical trials once they become pregnant. If all the nonclinical DART data were available earlier in the drug development process, there might be more acceptance for retaining pregnant individuals in clinical trials. Additionally, when pregnant patients are removed from clinical trials due to a lack of nonclinical DART data, an important opportunity is lost to obtain information on PK, pharmacodynamics or safety in pregnant individuals, or to understand differences in these parameters during different stages of pregnancy. Therefore, in order to include more information in a product labeling about the use of a drug during pregnancy, there will need to be a change in the current testing paradigm such that pregnant individuals can be retained or enrolled into clinical trials, and that will need to be supported by a change in how we think about the need for and the timing of nonclinical DART data during the drug development period.

3 | Nonclinical Study Considerations for Nonmandatory Contraception Requirements in Clinical Trials and/or Retaining Individuals Who Become Pregnant in Clinical Trials

An optimized clinical development plan, where pregnant individuals are not excluded from participation in preauthorization clinical trials, would benefit from a pragmatic DART strategy. Where appropriate, DART studies would contribute to the inflection point, as described earlier, and contraception requirements could be made nonmandatory, and if pregnancy occurred the participant could be offered the opportunity to remain in the trial while receiving the study medication. In this scenario when a pregnancy is detected while participating in a clinical trial, the individual will be reconsented and provided with all known information (information from DART studies, target information, data from other compounds in the class) and any inherent risk of the untreated disease on pregnancy, along with the potential

therapeutic benefit the study participant is currently receiving. This is the future state we would want to achieve by accelerating pregnancy-related data collection during the clinical development process. However, in the current regulatory paradigm under ICH M3(R2) if a clinical trial participant were to continue to receive the study medication while pregnant the expectation is that all female DART studies (i.e., female FEED, EFD and PPND) should be completed, regardless of how long the exposure to the study medication is expected to occur. Therefore, a more pragmatic DART strategy is needed to support accelerated clinical development. By considering factors that include the nature of the target indication (e.g., chronic vs. acute condition) and pharmacokinetic considerations of the drug (e.g., short half-life vs. long acting), DART studies needed to support retention or enrollment of pregnant individuals could evolve from the current ICH M3 requirements (Table 1).

In clinical trials for acute conditions (where treatment duration would typically not exceed 2 months) with study medication that has a short half-life, removal of mandatory contraception would result in exposure during the first 3 months of pregnancy, and likely no exposure during the fetal period. For such a clinical scenario, FEED and EFD studies may be sufficient to provide an assessment of risk to pregnancy exposure, provided the WoE suggests a low risk for potential latent effects evident only postnatally (based on knowledge of the target, off-target effects, and results of other nonclinical safety studies). This approach could thereby accelerate data collection in pregnant individuals in clinical trials. Furthermore, PPND studies will need to be conducted for market authorization, thereby providing the complete DART package for wider use once authorized. Clinical trials evaluating treatment of acute infectious diseases such as malaria, bacterial and/or viral infection may benefit from this scenario where periods of exposure are limited. However, these acute conditions do also occur later in pregnancy and therefore pregnant patient populations need to be included in trials eventually. To support pregnant individuals in later stages of pregnancy, PPND studies need to be conducted, preferably before Phase 3 initiation. This approach would facilitate the inclusion of these patients in Phase 3 clinical trials (see also Section 4 for alternate study designs that could enable enrollment of pregnant participants in later stages of pregnancy). An obvious added advantage with this strategy is that early clinical experience with pregnant individuals would subsequently facilitate recruitment of pregnant individuals in a large Phase 3 program.

In clinical trials for chronic conditions or diseases where treatment/chemoprevention needs to be given every year, where treatment interruption is not appropriate and/or when the study medication has a long half-life, it may not be possible to limit pregnancy exposure to the first few months of pregnancy. For these chronic conditions, the prospective decision to remove mandatory contraception would need to consider the fact that there will be a need for study medication throughout the pregnancy for effective disease control. And, regardless of the kinetics of the study medication, continued treatment to control the disease throughout pregnancy would be essential for the health of the pregnancy (e.g., systemic lupus erythematosus) as many chronic diseases are also associated with adverse pregnancy outcomes when poorly controlled. Therefore, in this scenario, before making the decision to remove mandatory contraception

TABLE 1 | Pragmatic DART strategies to support nonmandatory contraception requirements and retaining pregnant individuals in clinical trials.

	Clinical development scenario	Molecule characteristics	DART study type and information needed for inflection point decision making
1	Acute dosing period—pregnancy exposure restricted to 1st and/or early 2nd trimester	Short half-life molecules, (NCEs) or a limited duration exposure	FEED, rodent, and nonrodent EFD (for NCEs); Target information and expected minimal 1st trimester exposure (for mAbs)
2 ^a	Acute dosing period—long-term pregnancy exposure	Long half-life molecules where exposure or PD effects expected to last during majority of pregnancy	FEED, rodent, and nonrodent EFD and PPND (NCE); NHP PPND if safety during pregnancy not known (mAbs)
3	Exposure occurs only after pregnancy detection (e.g., pregnancy PK, etc.)	N/A	EFD and PPND ^b
4 ^a	Chronic treatment needed for clinical efficacy	N/A	FEED, EFD, and PPND
5	Biologic directed at a foreign target such as bacteria, viruses, or parasites for acute or chronic treatment	N/A	None required. However, an assessment for potential binding to fetal proteins may be desirable

^aConduct of standard PPND could be delayed by use of adapted rodent EFD and nonrodent EFD or adapted PPND and rodent and nonrodent EFDs (see Section 4; Table 3).

^bMay not be needed for clinical trials with 1st trimester only exposures, but for trials with 2nd or 3rd trimester exposures, PPND study is needed.

requirements in clinical trials, all female DART studies (FEED, EFD and PPND) will likely be needed to enable a comprehensive benefit risk assessment. Alternatively, adapted EFD studies that include elements of a PPND study or “preliminary” PPND study instead of a full PPND study (see Section 5) may be sufficient to inform risks of maintaining treatment for WOCBP that become pregnant during a trial and to support preauthorization trials with nonmandatory contraception requirements.

4 | Nonclinical Study Design Adaptations for Future Considerations

While the full set of female DART studies, as outlined in ICH M3(R2) would enable reaching an inflection point in drug development that would support enrollment of pregnant individuals as well as removal of female contraception requirements, alternative approaches need to be considered in DART study designs if we are to accelerate the timeline for achieving this important milestone. The current EFD study designs usually dose pregnant animals from start of implantation to end of organogenesis, and fetuses are evaluated for developmental toxicity at the end of pregnancy. Normally, there is no dosing during the fetal period, while a PPND study enables an assessment of potential risks for fetal development that would not be captured solely by a standard EFD study. Furthermore, depending on the species the fetal period can vary, and when it comes to human pregnancy exposures, the fetal period comprises the major period of pregnancy (Table 2). Therefore, a standard EFD in rodents or rabbits does not capture a large percentage of the period of human pregnancy exposure.

For a more progressive DART strategy that accelerates clinical trials in pregnant individuals two options are proposed. The expectation would be that these options would be in one species (e.g., rodent) and would also need a second species in the case of

EFDs. For example, in practice one would conduct an adapted EFD in rodent and standard EFD in nonrodent.

Option 1 (adapted EFD): Adapt standard EFD study to obtain early indication of potential risks for pregnancy exposure during both the embryonic and fetal development period. An adapted EFD study would have a subset of animals (e.g., $n = 8$; in addition to the standard group sizes for the EFD segment of the study) that continue to receive study drug through the fetal period into the first week of lactation with a limited set of endpoints being evaluated.

Option 2 (preliminary PPND): Preliminary PPND study to obtain early information on potential risks for pregnancy exposure during embryonic and fetal development periods. This study would have a limited number of pregnant females (e.g., $n = 8$) dosed from gestation Day 6 to lactation Day 7 (rodent) and the limited set of endpoints evaluated.

Both studies would have limited endpoints evaluated within the first week of birth. Based on historical experience, survival status in rodents within the first week of birth is a good indication of the toxicity experienced during the in utero and peri-natal period. The limited set of endpoints that are considered relevant include parturition observations, external examination, mortalities, litter size, live/dead status, sex, and body weights.

Data arising from these studies would provide hazard assessment of a full pregnancy exposure and could be part of a WoE to provide preliminary information in determining the risk for pregnancy exposure versus the potential clinical benefit for individuals, i.e., risk-based considerations for the inflection point. If found acceptable by regulatory authorities, either of these alternative approaches could conceivably support the removal of mandatory contraception requirements in clinical trials and retention of individuals who become pregnant in a clinical trial

TABLE 2 | Comparative phases of gestation.

Species	Implantation	End of organogenesis (duration)	Fetal period (approximate % of gestation)	Parturition	References
Mouse	GD 5	GD 15 (10 days)	4 days (21%)	GD 19–20	DeSesso (2012)
Rat	GD 6	GD 17 (12 days)	4–5 days (22%)	GD 21–22	DeSesso (2012)
Rabbit	GD 7	GD 19 (13 days)	13 days (42%)	GD 30–32	DeSesso (2012)
Minipig	GD 11	GD 35 (24 days)	79 days (69%)	GD 114	Bode et al. (2010)
NHP	GD 9–11	GD 44–45 (~35 days)	~121 days (73%)	GD 166	DeSesso (2012)
Human	GD 6–13	GD 56 (~6–7 weeks)	~30 weeks (79%)	~GW 38	DeSesso (2012)

TABLE 3 | Adapted DART study application scenarios for early enrollment of pregnant individuals.

Clinical trial population	Size of trials	DART study package
For removal of mandatory contraception requirements and/or the continuation of trial participation for individuals who become pregnant during studies for nonobstetric indications, where dosing begins prior to pregnancy detection and continues throughout pregnancy	Trials with limited number of WOCBP (e.g., < 150)	FEED Nonrodent EFD Rodent adapted EFD OR—FEED, EFD in two species and rodent preliminary PPND
For pharmaceutical with a potential for teratogenicity but with potential benefit for WOCBP For treatment of indications where use in 1st trimester is avoided for safety concerns and while use during 2nd and 3rd trimesters can still improve maternal health benefiting neonatal/infant outcomes	Not applicable	EFD in two species; PPND study in rodents with exposure after organogenesis to weaning and ICH S5(R3) list of endpoints
Obstetric indications that occur during 2nd or 3rd trimester (preterm labor, preeclampsia)	Not applicable	Combined EFD and PPND rodent EFD in nonrodent
Parturition-related treatments	Not applicable	Rodent PPND or NHP ePPND

to continue getting the benefit from study medication. These types of adapted studies could support removal of contraception requirements and retention of WOCBP who become pregnant in late Phase 2b type studies or early Phase 3 studies where limited number (e.g., ≤ 150) of WOCBP participants are enrolled. It is proposed that these alternative approaches could be used to support early enrollment of pregnant women in clinical development scenarios for chronic indications and/or trials that include long-acting therapeutics with a limited number of pregnant individuals. Using an adapted EFD or preliminary PPND study to enable enrollment of pregnant individuals into trials could be considered an evolution of the DART strategy in drug development similar to when DART requirements evolved to using an abbreviated EFD study (a.k.a. preliminary EFD study; ICH M3(R2)) in lieu of a full GLP EFD study to allow the enrollment of WOCBP in clinical trials. Such modifications could be especially valuable when the disease left untreated is expected to

have a significant impact on pregnancy outcomes, and current available safe treatment options are limited or absent. However, it should be noted that for registration or when pregnancy exposures are expected to occur in a larger number of individuals in a clinical trial, full PPND studies are expected (Table 3).

When developing therapeutics for specific obstetric conditions (e.g., preterm labor, preeclampsia), the studies needed should be adapted based on the intended usage of the pharmaceutical. For example, a combined EFD and PPND study in one species along with an EFD study in a second species could be used to assess the relevant stages of development for pharmaceuticals developed to treat conditions that only occur during pregnancy and which will never be used in other patient populations. For pharmaceuticals that will only be given around the time of parturition (e.g., to induce the onset of labor), a single DART study could be designed to cover this stage of reproduction. For

pharmaceuticals intended for women during the 2nd or 3rd trimester of pregnancy, but not during labor, extending the dose period in the EFD studies through the fetal period and a WoE suggesting low risk for latent effects may negate the need to conduct a PPND study prior to evaluation of this population in clinical trials. Further, for drugs with a developmental risk related to exposure during organogenesis that has intended clinical use in late pregnancy, studies evaluating pregnant nonclinical species after the completion of organogenesis can be considered to characterize potential risks in the fetal period (Table 3).

5 | Other Considerations to Support Enrollment of Pregnant Individuals

5.1 | Use of Physiologically Based Pharmacokinetics (PBPK) Models to Support Enrollment

Physiological changes that occur during pregnancy (changes to body weight and body fat composition, prolonged gastrointestinal transit time, increases in body water, cardiac output, blood flow, plasma volume, glomerular filtration rate, etc.) can alter drug absorption, distribution, metabolism and excretion characteristics thereby necessitating dose adjustments to maintain therapeutic benefit and to avoid potential adverse effects. There are numerous examples of medicines, including some biologics, needing dose adjustments during pregnancy (ampicillin, methadone, lamotrigine, infliximab); (Picardo and Seow 2020; Pinheiro and Stika 2020). However, a survey conducted by IQ DruSafe (Andrews et al. 2019) found that there was less than a threefold difference in exposure between pregnant and non-pregnant animals. As such, this may limit the use of pregnant animal data to estimate the potential drug concentrations during human pregnancy.

Physiologically based pharmacokinetics (PBPK) models of increasing sophistication have also been developed over recent years. These models include various physicochemical properties of the molecule, anatomical and physiological features of whole animals or specific tissues and provide a means to translate animal data to humans. PBPK modeling based upon in vitro to in vivo extrapolation (IVIVE) approaches using data derived from human cells/tissues is considered a highly relevant approach when applied to pregnant women (Ke et al. 2018). PBPK modeling can also be used to predict the extent and rate of placental transport of drugs and to predict exposure through breast milk (Ke et al. 2018). These types of modeling along with non-clinical in vivo data could help to characterize potential risks related to pregnancy exposure. For example, if modeling indicates a reduced concern for fetal harm due to expected exposure levels being very minimal, this could alleviate safety concerns for conducting clinical trials in pregnant individuals.

5.2 | Types of DART Studies Recommended for Clinical PK Trials in Pregnant Individuals

When a targeted clinical or bridging PK trial in pregnant individuals is being considered, the type of DART nonclinical package should depend on the trimester of exposure. For a 2nd

trimester PK trial, EFD and PPND studies along with an assessment of genetic toxicity could be considered sufficient. As the FEED study evaluates the impact on early embryonic development to implantation, one could argue that this study is less important when enrolling individuals who are already pregnant. Likewise, drugs used late in pregnancy (3rd trimester) might not need FEED or EFD studies, but only a PPND study for support, as at this time in pregnancy organogenesis has been completed.

5.3 | Route of Administration Considerations When Enrolling Pregnant Individuals

In general, it is expected that the potential placental or lactational transfer of a drug will be related to the systemic exposure of the mother. Thus, the route of administration leading to systemic exposure is also important to consider when evaluating potential risk to the fetus or offspring. For example, for drugs administered via dermal, inhalation, intrathecal, intramuscular, subcutaneous or intraocular routes, it is critical to understand the systemic exposure in both nonclinical models and humans. When systemic exposure is negligible (e.g., levels not expected to have any biologically meaningful pharmacological effect), the risk to the fetus/offspring is also reduced. These routes of administration which provide very low systemic exposures are perfect candidates for including pregnant patients without the need for DART studies because the risk for adverse pregnancy outcomes is low.

6 | Nonclinical Studies Appropriate to Support Clinical Trials in Breastfeeding Individuals

The main purpose of a clinical study in breastfeeding individuals (lactation study) is to obtain information on the amount of drug that the nursing infant might be exposed to in order to inform potential safety concerns for the infant (US FDA 2019). Typically, this is measured or modeled as the amount of drug being transferred to milk, with secondary prediction of how much drug the infant is likely to imbibe from breastmilk, and in some cases, how much drug is systemically absorbed by the infant. These studies can also provide information on differences in exposure in lactating individuals compared with nonpregnant individuals, including how long it takes to reach nonpregnant/nonlactating exposure levels, as well as the effects of the medicinal product on milk production.

There are three main types of clinical lactation studies (US FDA 2019). The first two involve study of the lactating woman only who agree to stop breastfeeding. These types of studies are not associated with exposure of a breast-fed child to an investigational pharmaceutical. The primary endpoint is evaluation of the concentration of the medicinal product in the breastmilk or in the breastmilk and serum of the breastfeeding individual. These study types would be conducted in individuals who are not planning to breastfeed their infants during the course of the study, including cases where the drug is intended to be administered for a short amount of time and the milk can be collected, analyzed and then discarded. The measurement of medicinal product concentrations in serum and milk of the breastfeeding individual is used to calculate the daily infant dose and relative

infant dose, which is the weight-adjusted percentage of the maternal dose (Anderson and Momper 2020; US FDA 2019). This information supports an initial infant risk assessment but, in this study design, the infant is not exposed to the drug and an assumption is made that 100% of the drug is systemically absorbed by the infant. The third type of clinical study (mother-infant pair study) is where the breastfeeding individual continues to nurse the child during the study, and the primary outcomes include a PK assessment of the medicinal product in the serum and breast milk of the breastfeeding individual, as well as PK and safety outcomes in the breastfeeding infant.

The nonclinical studies that could be useful in supporting these lactational studies fall into two main categories: (1) studies that help to assess the risk for adverse effects in the breastfed infant and (2) studies that could help predict drug concentrations in human milk, as further described in the sections below.

6.1 | Studies to Support Risk Assessment in Breastfed Infants

Nonclinical studies that may be needed to support infant safety in a clinical lactation study would depend not only on the type of lactation study that is planned, but also on the duration of the study, whether the lactating mother is actively breastfeeding her infant, and whether the drug has been registered/marketed or is still in clinical development. For single dose or short duration lactation studies, breastfeeding can be suspended during the study, with breastmilk being temporarily pumped and discarded or stored for use by the infant while the mother has exposure to the medicine. For this type of design, where there is no infant exposure, nonclinical studies that evaluate effects on nursing offspring, i.e., PPND studies, is not required to enable the conduct of human lactation studies. Additionally, because the clinical trial study participant is no longer pregnant, EFD studies are not required to support the clinical study. These types of lactation studies are, therefore, similar to a human PK study and only standard non-DART nonclinical studies should be needed to support such clinical studies.

For clinical trials where the objectives include mother-infant pairs or where breastfeeding is not temporarily discontinued, it is important to evaluate the effects of a relevant exposure of the pharmaceutical in the offspring in a nonclinical study prior to conducting this type of study. A nonclinical study that could provide this information is the PPND study, where dams are dosed with the pharmaceutical throughout pregnancy and during lactation, the offspring are potentially exposed to the pharmaceutical through nursing, and detailed information on the growth and development of offspring is generated. This study does not typically include confirmation of drug exposure in the offspring, but this information may be helpful in interpreting exposure-related effects observed in the offspring. However, it should be noted that since this study involves dosing through both gestation and lactation, it can be difficult to determine the relative contribution of lactational exposure alone to any toxicity. Although dedicated lactational exposure studies (e.g., cross-fostering pups to new mothers during lactation) are not typically conducted, such a study could be considered in specific situations where this distinction is critical. It should also

be noted that milk composition and mechanisms of drug transport into milk are highly species-specific (Ventrella et al. 2021) and studies evaluating drug concentrations in animal milk are generally not reliable for estimating human milk concentrations (see below). Therefore, evaluating milk concentrations in rat PPND studies is not recommended.

Another potential source of nonclinical information relevant to infant safety would be a juvenile animal study (JAS), if conducted. These studies have a primary objective of assessing postnatal developmental toxicity of medicinal products intended to be used in pediatric populations. A JAS study is not always required during drug development and the ICH S11 guideline (ICH 2020b) should be used to determine whether this study is needed for this purpose. However, if a JAS has been conducted, and includes preweaning exposures in offspring, results from this study could also be used as part of a risk assessment for breastfed infants. Of note, in this case the exposure occurs via direct administration to the juvenile animal, not via lactational transfer; therefore, studies that use oral administration are likely to be most relevant, and the relevant exposure levels would need to be compared to the measured or predicted human infant exposure via breastmilk.

Information from a standard clinical lactation study evaluating drug concentrations in milk can be used to estimate a daily infant oral dose, but it does not provide any information on how much of that infant dose is systemically absorbed. Therefore, nonclinical information on bioavailability is also very important for assessing risk in the breastfed infant. For example, many small molecules and biopharmaceuticals are administered by parenteral routes of administration (s.c. or i.v.) because they have low oral bioavailability. Even if those therapeutics are present in milk and result in a substantial infant dose, many may not be expected to be absorbed in an infant and, therefore, the risk for systemic exposure and adverse effects in the infant could be low. Another consideration is that oral bioavailability in young infants could be quite different to adults; these considerations also need to be taken into account in any risk assessment.

For protein-based biopharmaceuticals, evaluating the systemic exposure in the offspring has been a typical approach for evaluating lactational transfer, and this approach has shown that oral uptake of biopharmaceuticals, particularly mAbs, varies depending on the postnatal period and species (Baker et al. 2025; Van de Perre 2003). In primates, the lactational transfer of mAbs is most important during the immediate postnatal period of colostrum production, as this is a period during which maternal antibodies would provide protection to a newborn infant (Stratigakis et al. 2023), and the GI tract is transiently permeable for the uptake of intact mAbs during this perinatal period. However, as early as 1–2 weeks postnatally, protein therapeutics, even if present in milk, are likely to be digested and are unlikely to be absorbed in an active form (Anderson 2021; Krysko et al. 2023). In contrast, absorption of intact proteins in rodents extends further into the postnatal period and, therefore, rodent studies are likely to overpredict exposure of offspring to monoclonal antibodies during lactation (Downes 2018; Neal-Kluever et al. 2019). When conducted, clinical studies also have shown that mAb-based therapeutics are poorly or not absorbed by breastfeeding infants (Anderson 2021; Ciplea et al. 2020; National Institute of Child

Health and Human Development 2006; Schwake et al. 2024; Torkildsen et al. 2022; Witt et al. 2025). This information about the general characteristics of lactational transfer for mAbs and other protein therapeutics can, thus, be used as part of the infant risk assessment for these molecules when planning clinical lactation studies with or without infant exposures.

6.2 | Estimation of Drug Concentrations in Milk

There are several factors that influence a medicinal product's concentration in breast milk, including its physicochemical properties, its mechanism of transport into milk, and the composition of the milk (Breitzka et al. 1997; Ito et al. 2013; Ito and Alcorn 2003). Therapeutics can enter breastmilk through passive diffusion, but there are also transporters that can actively pump drug into or out of milk. Therefore, milk to plasma concentration ratios can be close to one for drugs entering milk by passive diffusion, but greater than or less than one when active transport occurs (Ito et al. 2013). Different drugs have different mechanisms for entering breast milk and, importantly, these mechanisms are often not conserved across species for any given drug (Ito et al. 2013). Additionally, the composition of breast milk varies greatly across species, which can also influence the concentration of drug in breast milk (Ventrella et al. 2021).

Therefore, nonclinical studies that evaluate the transfer of drugs into milk are not considered relevant for estimating drug concentrations in human milk (Wang et al. 2017). For this reason, the FDA states in its Guidance to Industry for labeling (US FDA 2020) that if animal lactation data are the only available data, the labeling can state that the drug is present in animal milk, but that information cannot be used to estimate human milk concentrations or to calculate a human infant dose. This creates a data gap in new product labeling, as adequate information on human milk concentrations is not available and it is, therefore, not possible to estimate how much therapeutic would be imbibed by a nursing infant. Consequently, there is an opportunity for the development of more predictive models or assessments to better predict concentrations of therapeutics in human milk. In order to calculate both potential lactational and systemic exposure of breastfeeding infants, such models should also consider the uptake and/or elimination from the infant digestive tract.

The in vitro models developed uses either primary human mammary epithelial cells (hMEC) or primary Göttingen Minipig mammary epithelial cells (mpMEC) to simulate the human mammary epithelium that forms the main barrier between maternal blood and milk (Bernardini et al. 2024; La Mantia et al. 2024). This in vitro model was developed to measure bidirectional permeability coefficients of drugs across the blood-milk barrier. The primary human and Göttingen Minipig mammary epithelial cells formed tight monolayers that expressed known transport proteins consistent with an epithelial phenotype. Results from transport experiments demonstrated the ability of the hMECs to differentiate between low and high permeability compounds.

The in vivo model uses lactating Göttingen Minipigs to provide quantitative data, including milk/plasma ratios along

with milk and plasma concentration-time profiles in both the sow and piglets. The Göttingen Minipig was selected based on anatomy of the mammary glands, physiology of lactation and colostrum/milk composition, and practical aspects (Ventrella et al. 2021). Using this model, up to four drugs could be tested simultaneously, maximizing the use of the animals, and a population PK analysis (Huang et al. 2025) could be employed to identify the source of variabilities in the lactating postpartum Göttingen Minipigs. Four studies were conducted, each using a single drug, and one study was conducted using four (4) drugs simultaneously. In each study, blood concentrations from the sows and piglets and milk drug concentrations were collected. Milk/Plasma (M/P) ratios for the minipigs were determined and compared to known human M/P ratios for the same compounds and showed excellent correlation ($R^2=0.9017$). When the population PK analysis was conducted on data obtained in the in vivo study with amoxicillin, the PK model successfully captured the PK profiles in maternal plasma and milk. The model-based simulation indicated that the exposure of piglets to amoxicillin through breastfeeding was minimal, suggesting a low risk of adverse events. This conclusion was reinforced by the absence of adverse events in the piglets during the study.

Lactation PBPK models were developed to predict the concentration of drugs in human milk during the lactation phase and estimate the milk to plasma ratio of medicines excreted into human milk (Nauwelaerts et al. 2023). The models were developed using the Simcyp simulator version 21 (Simcyp Ltd., Sheffield, UK) and incorporated the in vitro blood milk barrier transfer rates determined by the in vitro models (see above) for predicting the extent of medicine transferred into human breast milk. Efforts are underway to extend this model to develop infant PBPK models. The lactation PBPK models resulted in adequate predictions of maternal plasma and milk concentration time profiles for the 11 drugs tested. This lactation PBPK model can be used to simulate the human milk concentration and calculate predicted infant doses for other similar drugs. Others have also reported the development of a similar lactation PBPK model (Abduljalil et al. 2017). This model incorporated mammary tissue volume and blood flow, milk pH and milk intake to better predict lactational exposures. As we gain experience, refinement of these models is likely to improve their utility; a robust in silico approach could eventually obviate the value in conducting whole animal, or even in vitro, testing to predict lactational transfer.

7 | Conclusions

In summary, the prevailing approach to nonclinical testing and drug development has long hinged on the assumption that pregnancy can be circumvented during the clinical development of new medicines. Yet, this paradigm fails to address the significant unmet medical needs of pregnant and lactating individuals, whose health concerns have been overlooked for too long. The emergence of the ICH E21 guideline underscores the urgent necessity for change, advocating for a more inclusive and proactive framework. To this end, adaptations to nonclinical testing strategies are needed to balance animal use and high attrition rates in drug development with the need to include pregnant

individuals in clinical trials. Advancing relevant DART assessments to earlier stages of development, refining study designs to target specific risks, and leveraging predictive in silico modeling and in vitro assays offer promising avenues to more thoroughly evaluate potential impacts on both the maternal–fetal and mother–infant dyad so that a more complete DART dataset is available to support enrollment of pregnant and/or lactating individuals during clinical trials. These innovations lay the groundwork for a future where the safety and well-being of all populations are considered at every step of drug development.

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Ethics Statement

The authors have nothing to report.

Conflicts of Interest

D. Stanislaus, C.J. Bowman, K.C. Brannen, S.N. Campion, A.M. DeLise, C. Demarta-Gatsi, B.P. Enright, W. Halpern, V. Sutherland, K. Thacker, K. Thompson, R.L. Wange, and T. White are employed by the pharmaceutical industry or consult with the pharmaceutical industry as stated in their respective affiliations. B. Tornesi and S. Chalon are employed by MMV, a not-for-profit organization. The authors declare that they have no known competing financial interests or personal relationships that could appear to influence the work reported in this paper.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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